ARJUNA

JEE 2026 BATCH

Chemistry

Chemical Equilibrium

Lecture No- 03

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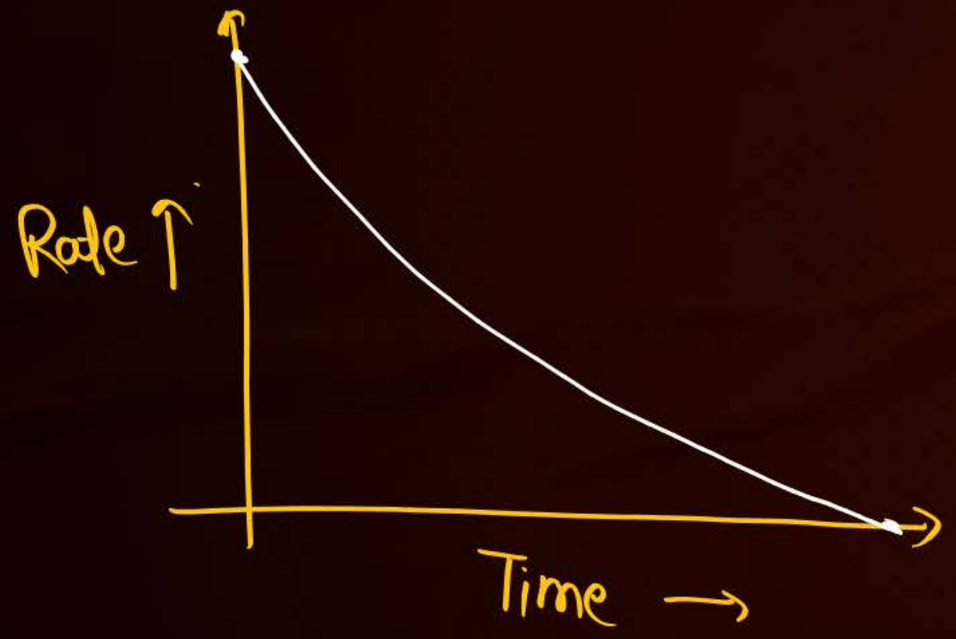
Topics to be covered

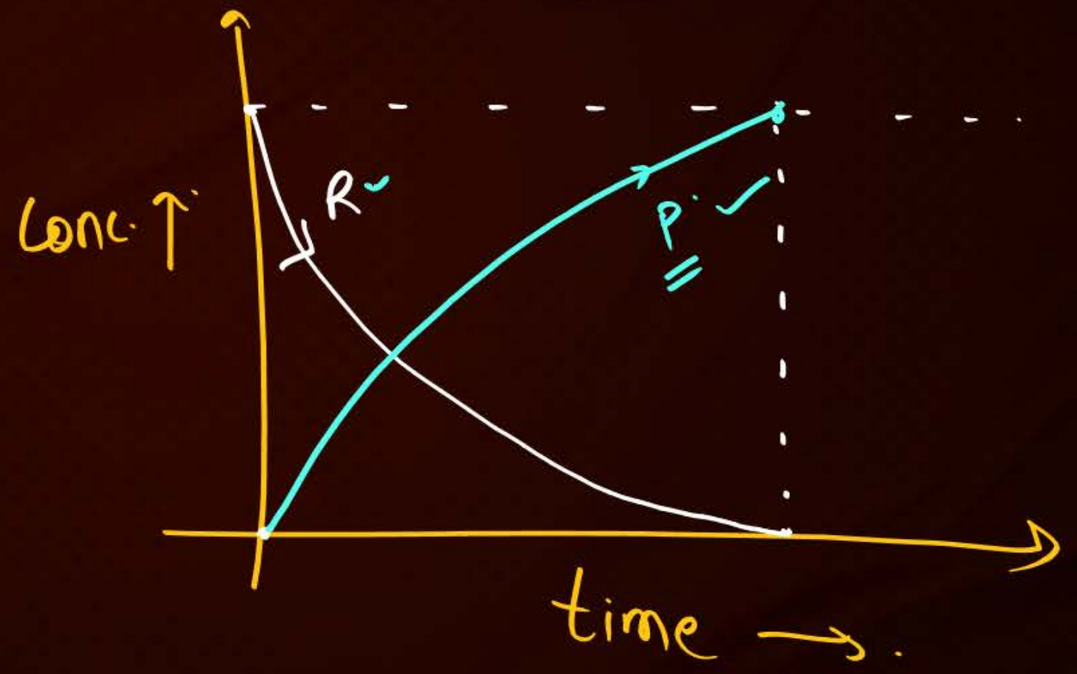
1. Chemical equilibrium
2. Equilibrium constant
3. K_p, K_c

Equilibrium in Chemical Process

Chemical Equilibrium

➢ **Graphical Representation:** [for Irreversible Rx^n]. $R \rightarrow P$





Chemical Equilibrium

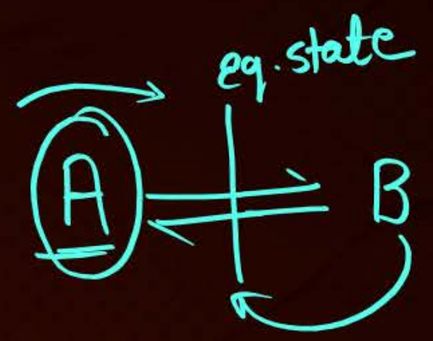
Characteristics of chemical equilibrium:

$R_f = R_b$

- Rate of forward reaction becomes equal to the rate of backward reaction.
- Concentration of all the reactants and products becomes constant.
- Dynamic in nature.
- Equilibrium is attained when none of the product is allowed to escape out or separate out from the reaction mixture.

ppt X

Chemical Equilibrium

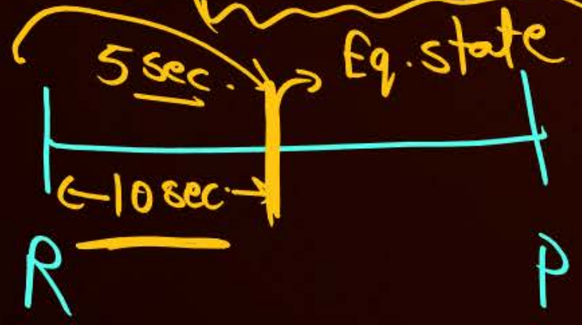


Characteristics of chemical equilibrium:

- Equilibrium can be attained from either directions, from the direction of reactants or products.

imp

- A catalyst does not alter the state of equilibrium. The only effect is that the equilibrium is attained quickly. It increases the rate of both forward and backward reaction to the same extent.



Chemical Equilibrium

Done

Law of Chemical Equilibrium and Equilibrium Constant:

$$aA_{(g)} + bB_{(g)} \rightleftharpoons cC_{(g)} + dD_{(g)}$$

$$R_f \propto [A]^a [B]^b$$

$$R_f = k_f [A]^a [B]^b$$

$$R_b \propto [C]^c [D]^d$$

$$R_b = k_b [C]^c [D]^d$$

At eq. $\rightarrow R_f = R_b$

$$k_f [A]^a [B]^b = k_b [C]^c [D]^d$$

$$\boxed{K_{eq} = \frac{k_f}{k_b} = \frac{[C]^c [D]^d}{[A]^a [B]^b}}$$

Law of chemical Eq.

Eq. constant.

Chemical Equilibrium

Done

Law of Chemical Equilibrium and Equilibrium Constant:

* $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

$$K_{eq} = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$

* $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

$$K_{eq} = \frac{[H_2][I_2]}{[HI]^2}$$

* $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$$K_{eq} = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

* Types of K_{eq} :→

- K_c
- K_p ← AA
- K_{pc}
- K_x

* K_c [Eq. Constant in terms of Concentration]

Done

* $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

$$K_c = K_{eq} = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$

* $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$$K_c = K_{eq} = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

* $C(s) + O_2(g) \rightleftharpoons CO_2(g)$

$$K_{eq} = \frac{[CO_2]}{[C][O_2]}$$

सब कुछ Include करना है

$$K_{eq} \cdot [C] = \frac{[CO_2]}{[O_2]}$$

$$K_c = \frac{[CO_2]}{[O_2]}$$

K_{eq} → सब कुछ आयेगा।

K_c → Pure solvent ✗
Pure liquid ✗
Pure solid ✗

K_c → Gas ✓
Solute (Aq.) ✓

* $A_{(g)} + C_{(s)} \rightleftharpoons B_{(g)}$

$$K_{eq} = \frac{[B]}{[A][C]}$$

$$K_c = \frac{[B]}{[A]}$$

Done

* $CH_{4(g)} + 2O_{2(g)} \rightleftharpoons CO_{2(g)} + 2H_2O_{(l)}$

$$K_{eq} = \frac{[CO_2][H_2O]^2}{[CH_4][O_2]^2}$$

$$K_c = \frac{[CO_2]}{[CH_4][O_2]^2}$$

* K_p [Eq. Constant in terms of Pressure]

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

$$K_{eq} = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$

$$K_p = \frac{(P_{PCl_3})(P_{Cl_2})}{(P_{PCl_5})}$$

P_{PCl_3} :- Partial Pressure of PCl_3

P_{Cl_2} :- ——————— of Cl_2

P_{PCl_5} :- ——————— of PCl_5

$P_{PCl_3} = X_{PCl_3} \cdot P_T$

* $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$$K_p = \frac{(P_{NH_3})^2}{(P_{N_2})(P_{H_2})^3}$$

Done

* $C(s) + O_2(g) \rightleftharpoons CO_2(g)$

$$K_p = \frac{P_{CO_2}}{P_{O_2}}$$

$K_p \not\rightarrow$ Pure liquid X

Pure solid X

* $CH_4(g) + 2O_2(g) \rightleftharpoons CO_2(g) + 2H_2O(l)$

$$K_p = \frac{P_{CO_2}}{(P_{CH_4})(P_{O_2})^2}$$

↳ K_p is defined for Rx^n in which atleast one of the reactant or Product is in gaseous state.

$$CH_3COOH_{(aq.)} + C_2H_5OH_{(aq.)} \rightleftharpoons CH_3COOC_2H_{5(aq.)} + H_2O_{(l)}.$$

↳ K_p is not defined.

↳ K_c is defined.

Done

Done

Chemical Equilibrium

Relationship between K_p and K_c

$$K_p = K_c [RT]^{\Delta n}$$

R :- Universal gas constant.

T :- Temp. (in kelvin).

$$\Delta n = \left[\begin{matrix}\text{No. of gaseous} \\ \text{moles of Products}\end{matrix}\right] - \left[\begin{matrix}\text{No. of gaseous} \\ \text{moles of Reactants}\end{matrix}\right]$$

* $PCl_{3(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

$\Delta n = (1+1) - (1) = 1$

* $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

$\Delta n = 2 - [1+3] = -2$

* $2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}$

$\Delta n = [1+1] - 2 = 0$

* $C_{(s)} + O_{2(g)} \rightleftharpoons CO_{2(g)}$

$\Delta n = 1 - [1+0] = 0$

* $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$

$\Delta n = 1 - 0 = 1$

$$* \quad aA(g) + bB(g) \rightleftharpoons cC(g) + dD(g)$$

$$K_P = \frac{(P_C)^c (P_D)^d}{(P_A)^a (P_B)^b}$$

for gases $\Rightarrow$ $PV = nRT$

$$P = \left(\frac{n}{V}\right) RT$$

$$P = C \cdot RT$$

$C \rightarrow$ concentration

$$K_P = \frac{\{[C]RT\}^c \{[D]RT\}^d}{\{[A]RT\}^a \{[B]RT\}^b}$$

$$K_P = \frac{[C]^c [D]^d}{[A]^a [B]^b} \frac{(RT)^{c+d}}{(RT)^{a+b}}$$

$$K_P = K_C \cdot (RT)^{(c+d)-(a+b)}$$

$$K_P = K_C (RT)^{\Delta n}$$

Done

Chemical Equilibrium

Units of K_p and K_c

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$

$$K_c = \frac{(\text{Conc.})^{1+1}}{(\text{Conc.})^{1}}$$

$$K_c = (\text{Conc.})^{(1+1)-1} \rightarrow \Delta n$$

$$K_c = (\text{Conc.})^{\Delta n}$$

* $K_c = [\text{Conc.}]^{\Delta n}$

$K_c = [\text{mol litre}^{-1}]^{\Delta n} \rightarrow (\text{mol m}^{-3})^{\Delta n}$

* $K_p = (\text{Pressure})^{\Delta n}$

$K_p = (\text{atm})^{\Delta n}$

$= (\text{Pascal})^{\Delta n}$

Done

$$K_p = K_c (RT)^{\Delta n}$$

T :- Temp. (in K)

Done

K_p	K_c	R
atm.	mol litre^{-1}	0.0821 atm $L K^{-1} mol^{-1}$ [1/12]
Pascal	mol m^{3-}	8.314 $J K^{-1} mol^{-1}$

NOTE :- Now a days K_p and K_c are considered as unit less.

$$K_p = K_c (RT)^{\Delta n}$$

Case (1) if $\Delta n = 0$

$K_p = K_c$

Case (2) if $\Delta n > 0$

$K_p > K_c$ $(RT > 1)$

$K_p < K_c$ $(RT < 1)$

Case (3) if $\Delta n < 0$

$K_p < K_c$ $(RT > 1)$

$K_p > K_c$ $(RT < 1)$

* $RT < 1$

$RT = 1$

$\frac{1}{12} \times T = 1$

$T = 12K$

$T > 12K$, $RT > 1$

$T < 12K$, $RT < 1$

Question

The following concentrations were obtained for the formation of NH_3 from N_2 and H_2 at equilibrium at 500 K. $[N_2] = 1.5 \times 10^{-2}$M. $[H_2] = 3.0 \times 10^{-2}$ M and $[NH_3] = 1.2 \times 10^{-2}$ M. Calculate equilibrium constant.

Ans:->

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{(1.2\times10^{-2})^2}{(1.5\times10^{-2})(3\times10^{-2})^3}$$

Question

$\Delta n = 3 - 2 = 1$

For the equilibrium, $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$ the value of the equilibrium constant, K_c is 3.75×10^{-6} at 1069 K. Calculate the K_p for the reaction at this temperature?

Ans :->

$K_c = 3.75 \times 10^{-6}$

$T = 1069\ K$

$K_p = ?$

$K_p = K_c (RT)^{\Delta n}$

$R = 0.0821$

Question

$\Delta n = 1 - \{1 + 1/2\} = -1/2$

For reaction: $SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$ [31 January-I Main-2023]

$K_p = 2 \times 10^{12}$ at 27°C and 1 atm pressure. The K_C for the same reaction is _____ $\times 10^{13}$. (Nearest integer)

(Given: R = 0.082 L atm $K^{-1}mol^{-1}$)

HW

Ans: $K_p = 2 \times 10^{12}$

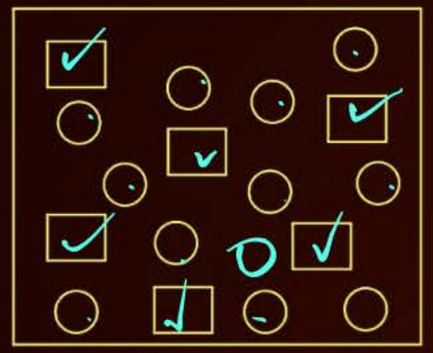
$T = 27°C = 300K$

$K_p = K_c(RT)^{\Delta n}$

Ans: [1]

Question

In the figure shown below reactant A (represented by square) is an equilibrium with product B (2020 Main, 9 Sep. II)



A → 6

O → 12

(represented by circle). The equilibrium constant is:

(A) 4

(B) 8

(C) 1

(D) 2

$K_{eq} = \frac{[P]}{[R]} = \frac{12}{6} = 2$

Question

HW

Consider the reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$. (2021 Main, 17 March II)

The temperature at which $K_C = 20.4$ and $K_p = 600.1$, is _____K. (Round off to the Nearest Integer).

[Assume all gases are ideal and $R = 0.0831$ L bar $K^{-1}mol^{-1}$]

$K_p = K_c (RT)^{\Delta n}$

Ans: [354]

Question

In which one of the following equilibria, $K_p \neq K_c$?

(2019 Main, 12 April II)

(A) $2C(s) + O_2(g) \rightleftharpoons 2CO(g)$

(B) $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

(C) $NO_2(g) + SO_2(g) \rightleftharpoons NO(g) + SO_3(g)$

(D) $2NO(g) \rightleftharpoons N_2(g) + O_2(g)$

$K_p = K_c (RT)^{\Delta n}$

If $\Delta n = 0$

$K_p = K_c$

Question

HW

For the reaction, $SO_2(g) + \frac{1}{2} O_2(g) \square SO_3(g)$

if $K_P = K_C (RT)^x$ where, the symbols have usual meaning, then the value of x is (assuming ideality) (2014 Main)

(A) -1

(B) $-\frac{1}{2}$

(C) $\frac{1}{2}$

(D) 1

Question

HW

For the reversible reaction,

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

at 500°C, the value of K_P is 1.44×10^{-5} when partial pressure is measured in atmosphere. The corresponding value of K_C with concentration in mol/L is

(2000, S, 1M)

(A) $\frac{1.44\times10^{-5}}{(0.082\times500)^{-2}}$

(B) $\frac{1.44\times10^{-5}}{(8.314\times773)^{-2}}$

(C) $\frac{1.44\times10^{-5}}{(0.082\times773)^{2}}$

(D) $\frac{1.44\times10^{-5}}{(0.082\times773)^{-2}}$

* K_{PC}. [Eq. constant in terms of Pressure and Concentration].

Done

Ex :→ $A_{(aq.)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$.

$$K_{PC} = \frac{(P_C)(P_D)}{(P_B)[A]}.$$

Ex :→ $HCl_{(g)} \rightleftharpoons H^+_{(aq.)} + Cl^-_{(aq.)}$.

$$K_{PC} = \frac{[H^+][Cl^-]}{P_{HCl}}.$$

NOTE :→ K_p is defined for Rx^n in which :-

(i) At least one of the Product or reactant should be in gaseous state.

(ii) None of the reactant or Product should be in aqueous state.

Done

* K_x. [Eq. Constant in terms of mol fraction].

$$A_{(g)} \rightleftharpoons B_{(g)} + C_{(g)}.$$

At eq:→ x_A x_B x_C.

$$K_x = \frac{x_B \cdot x_C}{x_A}.$$

Homogeneous Equilibria

Done

- In a homogeneous system, all the reactants and products are in the same phase.

$$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$$

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$$

$$CH_3COOH_{(aq.)} + C_2H_5OH_{(aq.)} \rightleftharpoons CH_3COOC_2H_{5(aq)} + H_2O$$

Heterogeneous Equilibria

- In a heterogeneous system, reactants and products are in two or more phases.

$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$$

$$C_{(s)} + O_{2(g)} \rightleftharpoons CO_{2(g)}$$

$$CH_{4(g)} + 2O_{2(g)} \rightleftharpoons CO_{2(g)} + 2H_2O_{(l)}$$

Heterogeneous Equilibria

Note:

In heterogeneous equilibrium, concentration of pure solid and pure liquid will not be included in the expression of K_p and K_c

Aaj ke liye
itna hi
Physics Wallah

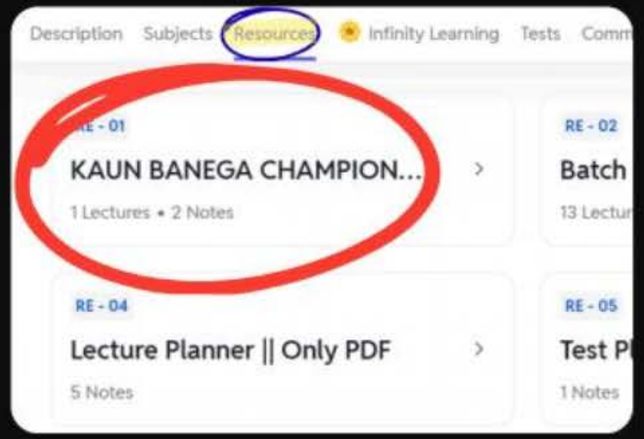
🎉 KBC Leaderboard is OUT !

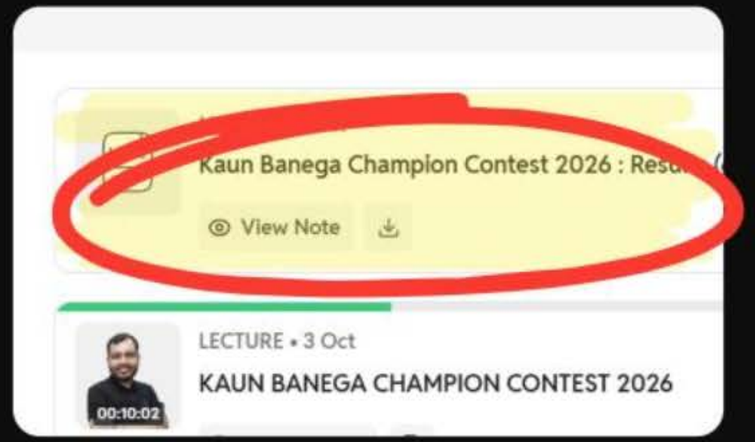
(21st Oct – 3 Nov)

Checkout

Check your batch "**Resources**" to see the PDF list of names of top students

Keep studying, **next leaderboard might have your name** shining at the top! 🚀





NOTE

Students found using unfair means during competition to increase XP have been excluded from the list

Keep Studying to Win

KBC Dates:

Until 31st Dec 2025

Watching Lectures (Live Or Recorded)

Solving DPPs

Attempting Weekly Tests

KAUN BANEGA CHAMPIONS

XP

Free iPhone

Free Laptop

Homework from Module

- DPP-1.
- Revision.
- जो class में दिया है.

THANK *You*